

Taking Dual Shortcut

Primal (Max)	Dual (min)
i th con $\leq$	i th var $\geq$
i th con $\geq$	i th var $\leq$
i th con $=$	i th var unrestricted
j th var $\geq$	j th con $\geq$
j th var $\leq$	j th con $\leq$
j th var unrestricted	j th con $=$

Note that the costs of the primal are now the b of the dual and that the b of the primal are now the costs of the dual.

When trying to figure out how taking the dual of parametric LP's think about when a variable appears in more than one constraint, in which case those constraints are all important for that variable in the dual. If a variable x appears in constraints $a_1, \dots, a_l$ then those constraint variables $a_1, \dots, a_l$ appear in the dual in the constraint corresponding to the variable x . Additionally, these constraint variables appear in this specific constraint with the same scalar multiple in which x appeared in the individual constraints.

Finding Basic Feasible Solutions

A bfs $\bar{x}$ is found by finding a basis B of A and inverting B then multiplying by b to get the vector x where any variables not in the basis b are simply set to 0.

$$\pi \ \& \ c^\pi$$

$$\text{For a basis } B \text{ of } A, \pi = (B^{-1})^T C_B$$

$$C^\pi = C - A^T \pi$$

Proving Optimality

Either show that $c^\pi \geq 0$ or that complementary slackness is 0 ie $y(Ax - b) = x(A^T y - c) = 0$. Note complementary slackness only applies to non equality constraints as the complementary slackness would be 0 for any feasible solution.

Taking the Dual of HW 2 Q5

$$\begin{aligned}
 & \min \sum y_h \\
 & \text{s.t. } x_{Ih} \leq y_h, (I \in A, h = 1, \dots, n) \\
 & \quad x_{Ih} + x_{Jh} \leq 1, ((I, J) \in c(A), h = 1, \dots, n) \\
 & \quad \sum_h x_{Ih} = 1, (I \in A) \\
 & \quad x_{Ih}, y_h \leq 1, (I \in A, h = 1, \dots, n)
 \end{aligned}$$

so

$$\begin{aligned}
 & \min \sum y_h \\
 & \text{s.t. } x_{Ih} - y_h \leq 0, (I \in A, h = 1, \dots, n) \\
 & \quad x_{Ih} + x_{Jh} \leq 1, ((I, J) \in c(A), h = 1, \dots, n) \\
 & \quad \sum_h x_{Ih} = 1, (I \in A) \\
 & \quad x_{Ih}, y_h \leq 1, (I \in A, h = 1, \dots, n) \\
 & \quad x_{Ih}, y_h \geq 0, (I \in A, h = 1, \dots, n)
 \end{aligned}$$

Dual

Lets denote the first constraints as our districts $d_{ij}, i \in A, j \in n$ and the next set as conflicts $\beta_{ikj}, (i, k) \in c(A), k \in n$, the next as appearances $\alpha_i, i \in A$ and lastly upper bounds as $ux_{Ih}, I \in A, h \in n$ and $uy_h, h \in n$

Thus our dual is:

$$\begin{aligned}
 & \max \sum \beta + \sum \alpha + \sum ux_{Ih} + \sum uy_h \\
 & \text{s.t. } d_{Ih} + \sum_{(I, J) \in c(A)} \beta_{IJh} + \alpha_I + ux_{Ih} \leq 0, (I \in A, h = 1, \dots, n) \\
 & \quad - \sum_{i \in A} d_{ij} + uy_j \leq 1, (j = 1, \dots, n) \\
 & \quad d \leq 0, \beta \leq 0, u \leq 0
 \end{aligned}$$